PIERO GASPAROTTO, PHD

Quantum Software Engineer @ Microsoft Quantum

@ piero.gasparotto@gmail.com Seattle, WA piero-gasparotto

EXPERIENCE

Quantum Software Engineer

Quantum Applications, Microsoft Quantum

Oct 2023 – Present Redmond, USA

I lead the development of end-to-end AI4Science solutions at the interface between machine-learning force fields, computational chemistry, quantum computing and cloud-native agentic workflows.

Principal ML Scientist

AI4Chemistry, Ro5

Mar 2023 – Oct 2023 New York, USA

Drove development of Ro5's 3D structure-based drug design platform and co-authored a peer-reviewed paper introducing Hydrascreen, a novel ML model for predicting protein-ligand binding affinities

Software Scientist

Scientific IT, Paul Scherrer Institute

May 2021 – Oct 2023 Villigen, CH

My work focused on developing and deploying machine-learning–based methods to provide real-time feedback during kHz serial X-ray protein crystallography experiments.

Postdoc

Empa – *nanotech@surfaces Group*

Apr 2019 – Mar 2021 Dübendorf, CH

Focused on ML Force Fields, enhanced-sampling workflows, super-hard alloys and modeling of surface-catalysis reactions.

Postdoc

University College London – *ICE Group*

Apr 2018 – Mar 2019 London, UK

Developed a novel machine-learning force field for carbon-based materials and studied glassy interfacial water properties by combining enhanced-sampling with molecular dynamics.

PhD Researcher

EPFL – *COSMO Group*

Oct 2013 – Mar 2018 Lausanne, CH

My research focused on unsupervised learning and pattern recognition methods combined with molecular dynamics and enhanced sampling techniques, with application to liquids, proteins and molecular crystals.

SELECTED PUBLICATIONS

- **QDK/Chemistry**: A Modular Toolkit for Quantum Chemistry Applications *arXiv preprint arXiv:2601.15253* (2026)
- **SimPoly**: Simulation of Polymers with Machine Learning Force Fields Derived from First Principles *arXiv preprint arXiv:2510.13696* (2025)

Full list: Google Scholar

EDUCATION

Ph.D. Computational Modelling

EPFL

2013–2018 Lausanne, CH

M.Sc. Materials Science

University of Padua

2009–2012 Padova, IT

B.Sc. Materials Science

University of Padua

2006–2009 Padova, IT

FUNDING

Torres-Quevedo Technological Transfer Grant (2019)
With Acellera Labs, for hybrid Quantum/ML approaches for modeling metallo-proteins (€98k)

HPC Compute Grants
Awarded 5 compute grants at CSCS and UKCP/MCC ($350k)

TECH SKILLS

Python | Scripting | Molecular Dynamics | Quantum Chemistry | MLFFs | CI/CD | Enhanced Sampling | Deep Learning | Cloud Computing | Distributed Systems | Scientific Software | Workflows Orchestration | Agentic AI | Customer-facing Applications

MANAGEMENT

Project Lead | Agile | Cross-functional Teams | Mentorship

CONTRIBUTOR

QDK/Chemistry — Microsoft Quantum Applications Toolkit
TORO-Indexer — A PyTorch indexer for SX crystallography
Hydrascreen — Protein-ligand binding affinities
PLUMED — Enhanced sampling for molecular dynamics
PAMM — Probabilistic analysis of molecular motifs

LANGUAGES

Italian
English
French